# Transforming Data to RDF: A Comparative Review of Tools, User Profiles and a Call for Community Contributions

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Abstract
Here we notify of ongoing work on a comparative review of Data transformation tools and frameworks. The review will focus on: flexibility of input and output formats and schemas, comparing interfaces, required expertise and decisions that are involved in the transformation.

**Keywords**
FAIR data, RDF mapping, semantic interoperability, ETL tools, community-driven review, knowledge graphs

## Introduction

Transforming heterogeneous datasets into standard or schema compliant RDF is essential for FAIR data practices in health and life sciences, yet remains a complex challenge. Numerous tools and frameworks such as RML [1], YARRRML[2], R2RML[3], RDFLib[4], RDFCraft[1], OpenRefine[5], Rightfield[6], FDP Populator[7], SeMPyRo[2], Ontop[8], etc. offer diverse approaches to this task, ranging from user-friendly interfaces to advanced scripting languages and virtual knowledge graph systems.

## Poster content

This poster presents the progress on an ongoing review of these tools, evaluating their flexibility, ease of use, and required expertise, while considering the human decision-making process involved in mapping data values to RDF concepts. We will also analyze the usage of these tools from the perspective of different roles, such as data owners, data stewards, and IT specialists.

To make this review comprehensive and community-driven, we invite SWAT4HCLS participants to contribute by sharing tools, frameworks, and experiences from their own fields.

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[1]https://github.com/MaastrichtU-IDS/RDFCraft
[2]https://github.com/Health-RI/SeMPyRO

These contributions will help build a living catalog of RDF mapping solutions, identify gaps, and highlight best practices for improving interoperability workflows in FAIR ecosystems.

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## Declaration on Generative AI

During the preparation of this work, the author(s) used Copilot order to: Improve text structure and clarity. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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